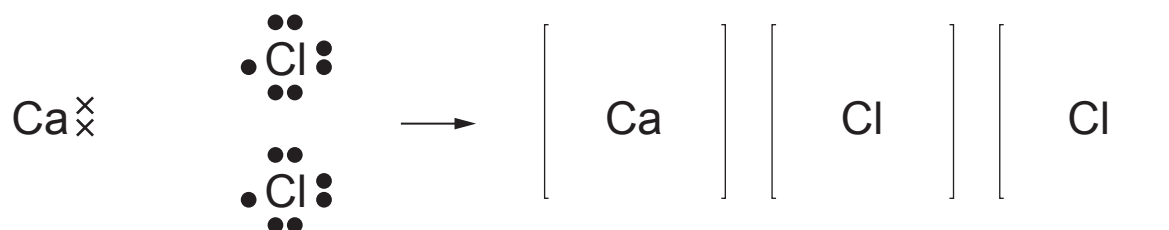


4. (a) Calcium reacts with chlorine to form the ionic compound calcium chloride.

Complete the dot and cross diagram to show the electronic changes that take place when calcium reacts with chlorine to form calcium chloride.

Show the charge on each ion.

[3]



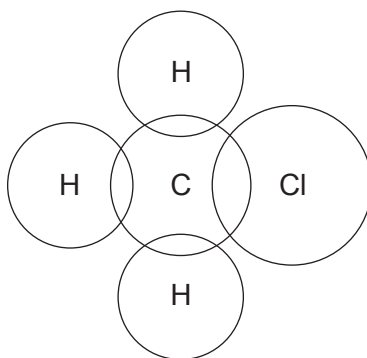
- (b) Complete the dot and cross diagram to show the bonding in a molecule of chloromethane.

[2]

hydrogen 1

carbon 2,4

chlorine 2,8,7



Examiner
only

(c) Calcium and calcium chloride have different bonding and structures.

Explain how each substance conducts electricity.

You may include diagrams in your answer.

[6 QER]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

11

